

## Sales Performance Analysis

### Project Objectives :

The objective of this project is to analyze sales data and identify key trends in product performance, regional sales, pricing, costs, and profitability.

The dataset contains 5,000 sales records.

### Tools Used :

SQL Server  
Microsoft Excel  
Power BI  
Python

### Business Questions :

1. How many sales transactions are in the dataset?
2. How many units were sold?
3. What is the average product price?
4. Which region has the most sales?
5. Which product has the most sales?

SQL Analysis

#### 1.Total Sales Transactions

Business Question :

How many sales transactions are in the dataset ?

SQL Query :

```
SELECT COUNT (*)  
FROM Sales_Import ;
```

Result :

5,000 transactions

Insight : The dataset contains 5,000 sales transactions.

#### 2.Total Units Sold

Business Question :

How many units were sold?

SQL Query :

```
SELECT SUM(Quantity)  
FROM Sales_Import ;
```

Result :  
14,908 units sold

Insight :  
A total of 14,908 product units were sold.

### **3.Average Product Price**

Business Question:  
What is the average product price?

SQL Query:

```
SELECT AVG(Price)
FROM Sales_Import;
```

Result:  
244.963

Insight:  
The average product price is approximately 244.96.

### **4. Sales by Region**

Business Question:  
Which region has the most sales?

SQL Query:

```
SELECT Region, COUNT(*)
FROM Sales_Import
GROUP BY Region;
```

Result:  
North: 1,039  
East: 997  
South: 1,534  
West: 1,430

Insight:  
The South region has the highest number of sales transactions with 1,534 transactions.

### **4. Sales by Region**

Business Question:  
Which region has the most sales?

SQL Query:

```
SELECT Region, COUNT(*)
FROM Sales_Import
GROUP BY Region;
```

Result:

North: 1,039

East: 997

South: 1,534

West: 1,430

Insight:

The South region has the highest number of sales transactions with 1,534 transactions.

## 5. Sales by Product

Business Question:

Which product has the most sales?

SQL Query:

```
SELECT Product, COUNT(*) AS Total_Sales
FROM Sales_Import
GROUP BY Product
ORDER BY Total_Sales DESC;
```

Result:

Monitor: 642 transactions

Insight:

Monitor has the highest number of sales transactions with 642 transactions.

## 6. Total Revenue

Business Question:

What is the total revenue generated from all sales?

SQL Query:

```
SELECT SUM(Quantity * Price) AS Total_Revenue
FROM Sales_Import;
```

Result:

3,636,282

Insight:

The company generated a total revenue of 3,636,282 from all sales transactions.

## 7. Total Profit

Business Question:

What is the total profit generated from all sales?

SQL Query:

```
SELECT SUM((Price - Cost) * Quantity) AS Total_Profit
FROM Sales_Import;
```

Result:

1,190,619

Insight:

The company generated a total profit of 1,190,619 from all sales transactions.

## 8. Profit by Product

Business Question:

Which product generates the highest profit?

SQL Query:

```
SELECT Product,  
       SUM((Price - Cost) * Quantity) AS Total_Profit  
FROM Sales_Import  
GROUP BY Product  
ORDER BY Total_Profit DESC;
```

Result:

Laptop: 459,991  
Desk: 197,279  
Monitor: 165,089  
Office Chair: 128,894  
Headphones: 80,055  
Webcam: 68,844  
Keyboard: 53,191  
Mouse: 37,276

Insight:

Laptop generated the highest total profit with 459,991, followed by Desk with 197,279. Mouse generated the lowest total profit with 37,276.

## 8. Profit by Product

Business Question:

Which product generates the highest profit?

SQL Query:

```
SELECT Product,  
       SUM((Price - Cost) * Quantity) AS Total_Profit  
FROM Sales_Import  
GROUP BY Product  
ORDER BY Total_Profit DESC;
```

Result:

Laptop: 459,991  
Desk: 197,279  
Monitor: 165,089  
Office Chair: 128,894  
Headphones: 80,055  
Webcam: 68,844

Keyboard: 53,191  
Mouse: 37,276

Insight:

Laptop generated the highest total profit with 459,991, followed by Desk with 197,279. Mouse generated the lowest total profit with 37,276.

## 9. Revenue by Region

Business Question:

Which region generates the highest revenue?

SQL Query:

```
SELECT Region,  
       SUM(Quantity * Price) AS Total_Revenue  
FROM Sales_Import  
GROUP BY Region  
ORDER BY Total_Revenue DESC;
```

Result:

South: 1,121,795  
West: 1,082,521  
North: 762,111  
East: 669,855

Insight:

The South region generated the highest revenue with 1,121,795, followed by the West region with 1,082,521. The East region generated the lowest revenue with 669,855.

## 10. Profit by Region

Business Question:

Which region generates the highest profit?

SQL Query:

```
SELECT Region,  
       SUM((Price - Cost) * Quantity) AS Total_Profit  
FROM Sales_Import  
GROUP BY Region  
ORDER BY Total_Profit DESC;
```

Result:

South: 367,549  
West: 352,397  
North: 247,848  
East: 222,825

Insight:

The South region generated the highest total profit with 367,549, followed by the West region with 352,397. The East region generated the lowest total profit with 222,825.

## 11. Profit Margin by Region

Business Question:

Which region has the highest profit margin?

SQL Query:

```
SELECT Region,

       SUM((Price - Cost) * Quantity) * 100.0

       / SUM(Price * Quantity) AS Profit_Margin

FROM Sales_Import

GROUP BY Region

ORDER BY Profit_Margin DESC;
```

Result:

East: 33.26%  
South: 32.76%  
West: 32.55%  
North: 32.52%

Insight:

The East region has the highest profit margin at approximately 33.26%. Although the South region generated the highest total revenue and profit, the East region achieved the highest profit margin.

## 12. Product Revenue Analysis

Business Question:

Which product generates the highest revenue?

SQL Query:

```
SELECT Product,

SUM(Quantity * Price) AS Total_Revenue

FROM Sales_Import

GROUP BY Product

ORDER BY Total_Revenue DESC;
```

Result:

Laptop: 1,661,169

Desk: 620,190

Monitor: 485,732

Office Chair: 382,604

Headphones: 162,162

Webcam: 147,490

Keyboard: 108,346

Mouse: 68,589

Insight:

Laptop generated the highest revenue, with total revenue of 1,661,169.